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MINI-PROJECT REPORT

COLLEGE PORTAL FOR STUDENTS

Submitted by

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BONAFIDE CERTIFICATE

RAJALAKSHMI ENGINEERING COLLEGE

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Certified that this Project Report titled "COLLEGE PORTAL FOR STUDENTS" is the Bonafide work of GANGA GOWRI. P (241801070) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Submitted to Project Viva-Voce Examination held on _______________

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ABSTRACT

The College Portal for Students is a frontend web application developed using HTML5, CSS3, and JavaScript that provides students of Rajalakshmi Engineering College with a centralized digital interface to access all essential academic information. The portal enables students to securely log in and view their attendance records, internal and examination marks, course materials, and institutional announcements from a single responsive platform.

The application features a personalised student dashboard displaying the student profile, current semester schedule, attendance summary, and recent announcements. A dedicated Attendance module presents subject-wise attendance percentages and eligibility status. The Marks module displays internal assessment scores and semester grade point averages. The Course Materials module allows students to browse and download lecture notes, lab manuals, and reference documents organised by subject. The Announcements module provides notices regarding examinations, events, holidays, and placement activities in a clearly structured feed.

The portal is implemented entirely on the client side without any backend infrastructure, making it lightweight and instantly deployable. The interface is fully responsive and accessible on desktop, tablet, and mobile devices. The Authentication module provides modal-based Sign In and Sign Up functionality for secure access. This project demonstrates the practical application of frontend web technologies in building a meaningful academic management tool for students.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
UX	User Experience
API	Application Programming Interface
DOM	Document Object Model
CDN	Content Delivery Network
HTTP	HyperText Transfer Protocol
ES6	ECMAScript 2015 – 6th Edition
LMS	Learning Management System
ERP	Enterprise Resource Planning
DB	Database
GPA	Grade Point Average
CGPA	Cumulative Grade Point Average
PDF	Portable Document Format
REC	Rajalakshmi Engineering College

CHAPTER 1

INTRODUCTION

1.1 General

Academic institutions today manage large volumes of student data including attendance records, examination results, course materials, and institutional announcements. Traditional methods of disseminating this information through physical notice boards, printed mark sheets, and manual attendance registers are inefficient, error-prone, and inaccessible to students outside the campus. With the rapid adoption of web technologies, there is a strong need for a dedicated digital portal that consolidates all essential student information into a single, accessible, and interactive platform.

The College Portal for Students is a frontend web application developed using HTML5, CSS3, and JavaScript for Rajalakshmi Engineering College (REC), Chennai. The portal provides students with a centralised interface to access their academic profile, attendance, marks, course materials, and notices without requiring any backend infrastructure. It is designed to be lightweight, responsive, and easy to use on any device.

1.2 Need for College Portal System

The need for a centralised College Portal arises from several challenges faced by students in accessing academic information:

- Absence of a unified platform where students can access attendance, marks, notes, and notices from one location.
- Difficulty in tracking subject-wise attendance and understanding eligibility status for examinations.
- Delay in accessing internal assessment results and grade reports.
- Inefficient distribution of lecture notes and study materials through physical means.
- Lack of a structured announcement system for timely communication of academic notices, holidays, and placement activities.

1.3 Scope of the System

The scope of the College Portal for Students includes a secure login and sign-up system, a personalised student dashboard with profile and schedule information, subject-wise attendance tracking with eligibility indication, internal and semester marks display with grade computation, course material browsing and download organised by subject, and an announcements feed with categorised notices. The portal is designed as a fully responsive single-page application compatible with all modern browsers and devices.

1.4 Organization of the Report

Chapter 1 introduces the project. Chapter 2 presents the literature survey of existing college management systems. Chapter 3 explains problem formulation and objectives. Chapter 4 describes system design and requirements. Chapter 5 discusses all implementation modules. Chapter 6 presents results and testing. Chapter 7 concludes the project with future enhancements.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

College management and student information systems have been widely studied in the domains of educational technology and enterprise resource planning. Several web-based and application-based platforms have been developed to facilitate digital academic administration. This chapter reviews existing college ERP and LMS systems and discusses related research that influenced the design of the College Portal for Students.

2.2 Existing College Management Systems

Platforms such as EduPrime, CampusCare, and Fedena offer comprehensive college ERP features including student enrollment, attendance tracking, grade management, and library management, but require full server-side deployment and database administration. Learning Management Systems such as Moodle and Google Classroom provide course content delivery and assignment submission features but are primarily designed for online learning rather than administrative data access. Most existing systems require institutional licenses, server hosting infrastructure, and administrative setup, making them inaccessible for lightweight academic projects. A browser-based, frontend-only portal addresses the need for instant deployment without backend complexity.

2.3 Related Work

Research on web-based student information systems highlights the importance of centralised data access in improving student engagement and academic planning. Studies on responsive educational portals demonstrate that mobile-compatible interfaces significantly increase student usage rates and satisfaction. Research on LMS adoption in Indian engineering colleges indicates that students prefer portals that provide quick access to attendance and marks without navigating complex administrative interfaces. These findings informed the decision to develop a focused, module-based portal addressing the most critical academic data needs.

2.4 Summary

Existing college management systems are either too complex for lightweight deployment or focus primarily on content delivery rather than student academic data access. There is a clear need for a simple, responsive, browser-based portal that provides engineering college students with quick and organised access to attendance records, marks, course materials, and institutional announcements. The College Portal for Students addresses this gap using frontend web technologies without requiring any backend infrastructure.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Students at engineering colleges frequently face difficulties in accessing their academic information in a timely and organised manner. The key problems identified are:

- No centralised system for students to access attendance, marks, course materials, and notices from a single platform.
- Difficulty in tracking subject-wise attendance and determining eligibility for end-semester examinations in real time.
- Delayed access to internal assessment marks and grade point averages.
- Inefficient distribution of lecture notes, lab manuals, and study materials through physical or fragmented digital channels.
- Absence of a structured announcement system, leading to students missing important notices regarding examinations, placements, and events.
- Most academic portals require backend infrastructure and institutional licensing, making them impractical for a frontend-only academic project.

3.2 Objectives of the System

The specific objectives of the College Portal for Students are:

- To provide a secure login and sign-up system for authenticated student access.
- To develop a personalised student dashboard displaying academic profile, semester schedule, and summary statistics.
- To implement a subject-wise attendance tracking module with percentage computation and eligibility indication.
- To display internal assessment and semester examination marks with grade calculation.
- To provide a course materials module for browsing and downloading subject-specific lecture notes and reference documents.
- To implement an announcements module with categorised notices covering examinations, placements, events, and holidays.
- To develop a fully responsive interface accessible on desktop, tablet, and mobile devices without any backend requirement.

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design defines the overall structure, components, and interaction flow of the College Portal. The portal is designed as a single-page web application with distinct section-based navigation. The design covers the frontend architecture, module layout, technology stack, and system requirements necessary for development and deployment.

4.2 System Architecture

The College Portal follows a three-layer client-side architecture.

Presentation Layer (HTML5):

Defines the complete structure using semantic HTML5 elements organised into sections: navigation bar, student dashboard, attendance section, marks section, course materials section, announcements section, and authentication modals.

Styling Layer (CSS3):

Handles all visual design using CSS variables, flexbox and grid layouts, keyframe animations, hover transitions, and media queries for responsive design across all screen sizes and devices.

Logic Layer (JavaScript):

Manages all dynamic functionality including tab switching between portal sections, data rendering for attendance and marks tables, modal handling for authentication, DOM manipulation for announcements and materials display, and scroll behaviour.

4.3 System Requirements

4.3.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	Minimum 2 GB
Storage	Minimum 100 MB free space
Display	Minimum resolution 1024 x 768
Internet	Required for CDN fonts and icons
Input Devices	Keyboard and Mouse or Touchscreen

Table 4.1: Hardware Requirements

4.3.2 Software Requirements

Component	Specification
Operating System	Windows 10, macOS, Linux, Android, iOS
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Frontend	HTML5, CSS3, JavaScript (ES6+)
Icon Library	Font Awesome 6.5 via CDN

Font Library	Google Fonts – Roboto, Open Sans
Development Editor	Visual Studio Code

Table 4.2: Software Requirements

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The College Portal for Students is implemented using a modular frontend development approach where each feature is an independent section within the single-page layout. Development followed a structured process: requirement analysis, UI wireframing, frontend coding, module integration, testing, and refinement. The three source files — index.html for structure, style.css for visual design, and script.js for dynamic logic — are maintained separately for clean code organisation. No backend or database is required, making the application lightweight and deployable on any standard web hosting environment.

5.2 Module Descriptions

The College Portal consists of seven major modules that together deliver a complete and seamless academic information experience for students.

5.2.1 Navigation Module

The Navigation Module provides the top navigation bar with the Rajalakshmi Engineering College logo and portal name, navigation links for Dashboard, Attendance, Marks, Materials, Announcements, and a Logout button. The navigation bar is fixed-position and transitions from transparent to a solid background with a shadow as the user scrolls. A hamburger menu is provided for mobile devices that toggles the navigation links using CSS transitions.

5.2.2 Student Dashboard Module

The Student Dashboard Module serves as the landing section of the portal after authentication. It displays a personalised welcome banner with the student’s name, roll number, department, and semester. Four summary cards present key statistics: overall attendance percentage, current CGPA, number of pending assignments, and total enrolled courses.

Below the summary cards, the dashboard is divided into two panels. The left panel displays a list of recent announcements with title, category tag, and a link to view the full notice. The right panel presents the student’s current-day timetable with subject codes, subject names, and scheduled time slots. This module provides an immediate academic overview without requiring navigation to individual sections.

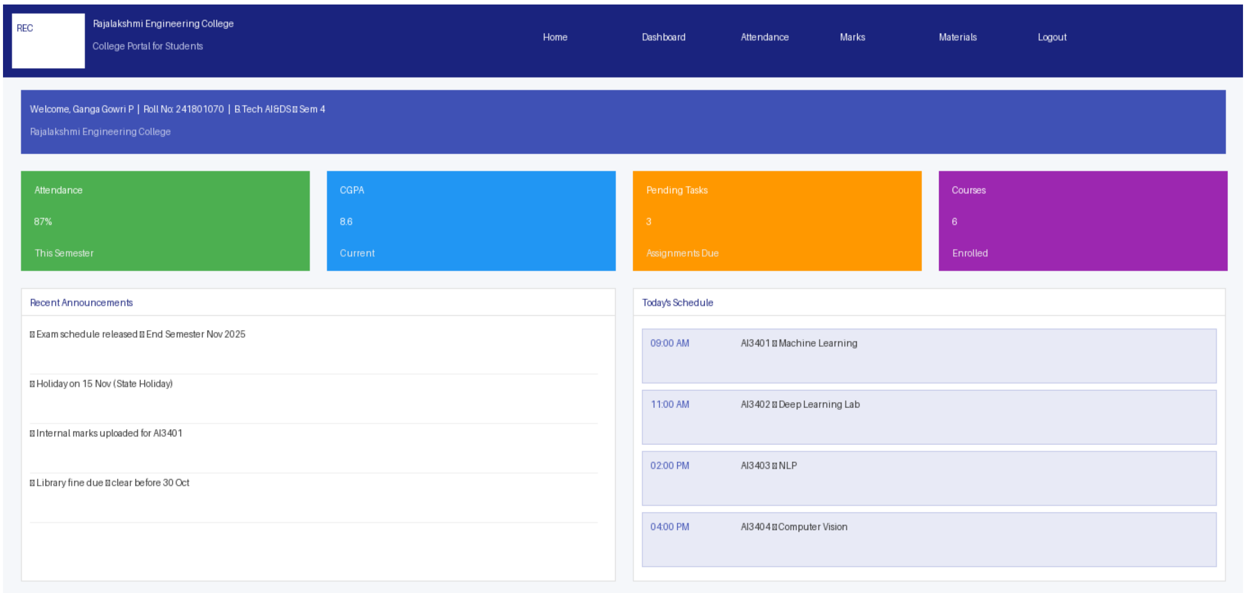

Figure 5.1: Student Dashboard

5.2.3 Course Materials Module

The Course Materials Module organises study resources by subject. Each subject is displayed as a card with the subject code, subject name, and the count of available resources. Within each subject card, individual resources are listed as downloadable items with the file name and a download icon. Resources include lecture notes, lab manuals, reference textbooks, and past examination papers in PDF format.

The module uses a grid layout that adapts from a four-column arrangement on desktop to a two-column arrangement on tablet and a single-column arrangement on mobile devices. Resource cards include hover animations that highlight the border and elevate the card for an engaging browsing experience.

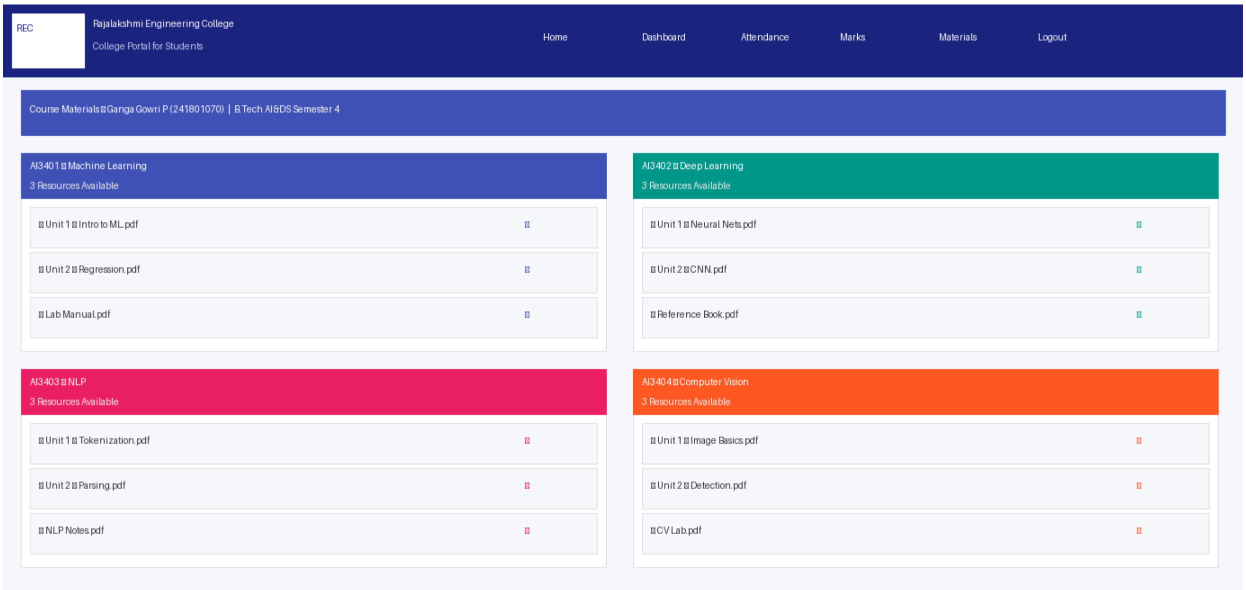

Figure 5.2: Course Materials Module

5.2.4 Attendance Module

The Attendance Module presents comprehensive attendance data for the current semester. At the top of the section, three summary cards display the overall attendance percentage, total classes attended out of total classes held, and the minimum attendance threshold required for examination eligibility. The overall attendance percentage is colour-coded in green when above 75 percent and in red when below the required threshold.

Below the summary cards, a detailed table lists each enrolled subject with the subject code, subject name, total classes held, classes attended, attendance percentage, and eligibility status. Each row is alternately shaded for readability. Students whose attendance falls below 75 percent are clearly indicated with a red status badge.

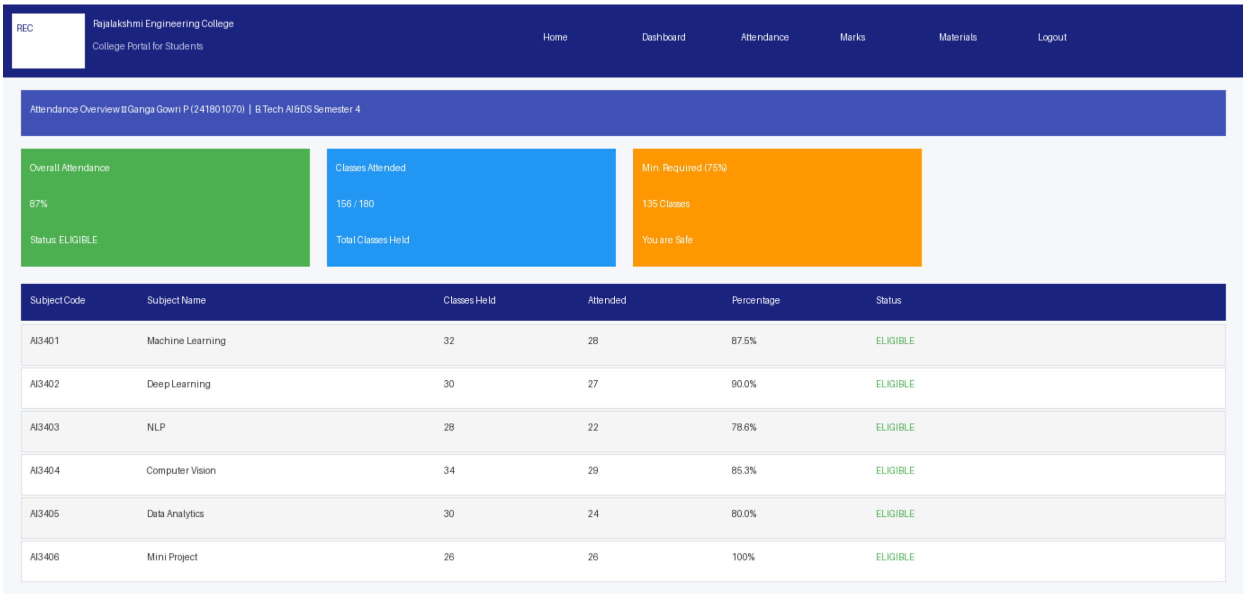

Figure 5.3: Attendance Module

5.2.5 Marks Module

The Marks Module displays assessment results for all enrolled subjects. Three summary cards at the top of the section present the current CGPA, the semester GPA for the most recent semester, and the student's departmental rank. These cards use distinct colour coding for quick identification.

A detailed internal assessment table lists each subject with the marks obtained in Internal Assessments 1, 2, and 3, the average of the best two assessments used for grade computation, and the resulting grade. A second table displays the end-semester examination marks for completed semesters with the semester GPA and result status. The module supports tab-based navigation to switch between internal marks and semester results.

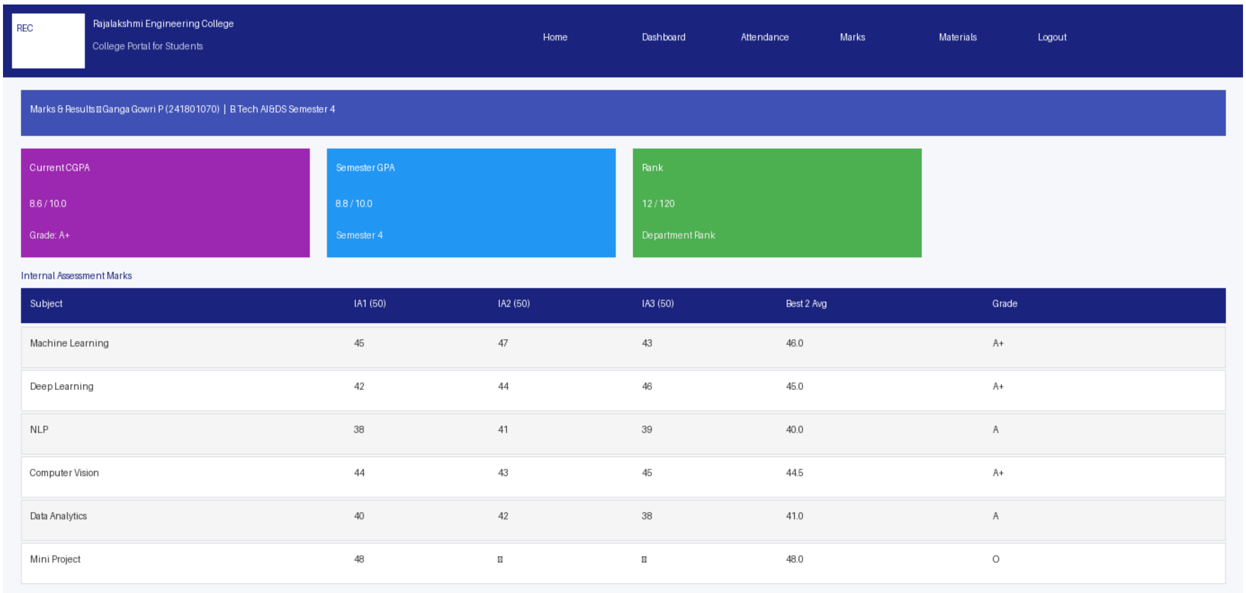

Figure 5.4: Marks Module

5.2.6 Announcements Module

The Announcements Module presents all institutional notices in a structured, categorised feed. Each announcement is displayed as a card with a coloured category badge on the left indicating the notice type: URGENT, NOTICE, INFO, or EVENT. The card displays the announcement title, a brief description, the posting date, and a View Details link.

Announcements are sorted with the most recent notices at the top. The category badge colours follow a consistent scheme: red for urgent notices, orange for general notices, blue for informational announcements, and green for events. This module ensures that students are always informed about upcoming examinations, placement drives, holidays, and college events.

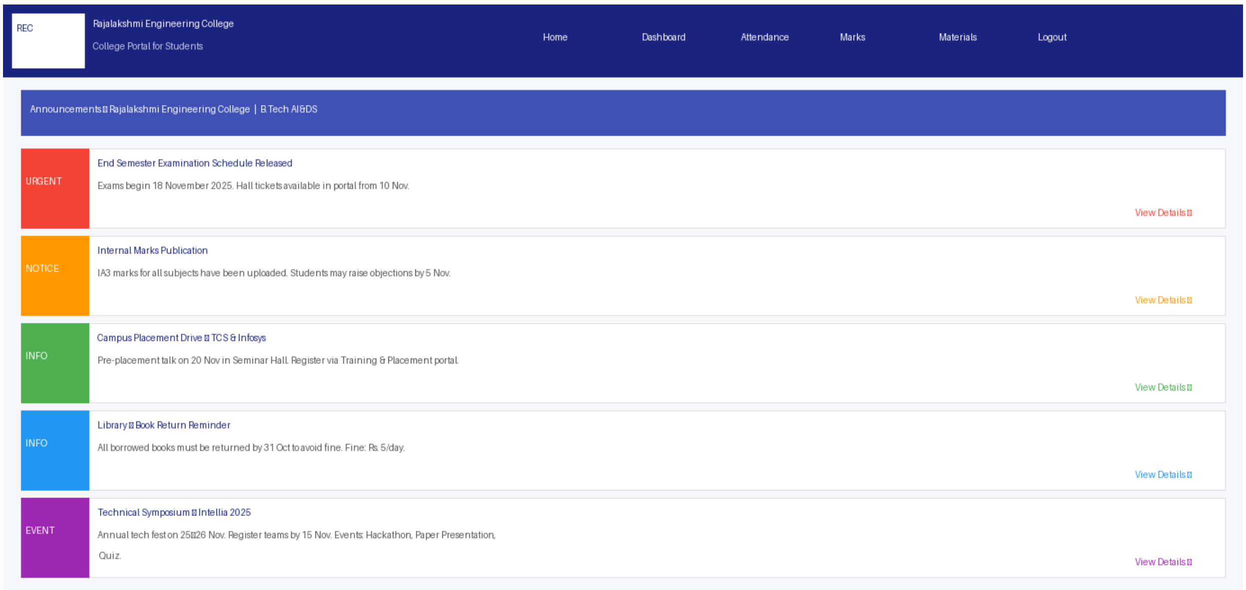

Figure 5.5: Announcements Module

5.2.7 Authentication Module

The Authentication Module provides modal-based Sign In and Sign Up interfaces accessible from the navigation bar. The Sign Up form collects the student's full name, roll number, email address, department, and password. The Sign In form collects the registered email and password. On successful submission, the showSuccess() function displays a confirmation message and redirects the user to the dashboard. The modal closes when the user clicks the close button, clicks outside the modal overlay, or presses the Escape key.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 System Output Screens

The College Portal for Students was successfully implemented and tested across Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari on mobile, tablet, and desktop devices. The following sections describe the key output screens of the portal.

Dashboard Screen

The student dashboard displays a personalised welcome banner with the student's name and academic details, four summary cards for attendance, CGPA, pending tasks, and enrolled courses, a recent announcements panel on the left, and the day's timetable on the right. The layout adapts cleanly to all screen sizes.

Attendance Screen

The attendance section presents the overall attendance percentage summary cards at the top followed by a detailed subject-wise attendance table. Eligible students are indicated with a green badge and students with attendance below the threshold are indicated with a red badge.

Marks Screen

The marks section displays the CGPA, semester GPA, and rank summary cards, followed by the internal assessment marks table with subject-wise scores and computed grades.

Course Materials Screen

The course materials section displays subject cards with downloadable resources. Each card presents the subject code, name, and individual resource items with download links.

Announcements Screen

The announcements section presents categorised notices with colour-coded tags, titles, and descriptions. Urgent notices, informational announcements, and event listings are clearly distinguished.

6.2 Performance Evaluation

The College Portal was evaluated on key performance parameters. The page loads in under two seconds on a standard internet connection. Attendance and marks tables render instantly on section navigation. The portal functions correctly on all major browsers. Animations run smoothly on modern browsers. Tab switching between portal sections is instantaneous and modal animations are fluid on all tested devices. The responsive layout adapts correctly across screen widths from 320 pixels to 2560 pixels.

6.3 System Testing

Manual black-box testing was performed to verify all portal features. The results are presented below.

S. No	Test Case	Expected Output	Actual Output	Status
1	Login with valid credentials	Dashboard loads with student data	As expected	PASS
2	Login with empty fields	Alert message shown	As expected	PASS

3	Dashboard summary cards	Attendance, CGPA, tasks displayed	As expected	PASS
4	Attendance table display	Subject-wise rows with percentages	As expected	PASS
5	Attendance eligibility badge	Green/Red based on percentage	As expected	PASS
6	Marks table display	IA scores and grades listed	As expected	PASS
7	Course materials grid	Subject cards with resource links	As expected	PASS
8	Announcements feed	Notices with category badges	As expected	PASS
9	Sign Up modal	Modal with registration form	As expected	PASS
10	Sign In modal	Modal with login form	As expected	PASS
11	Overlay click closes modal	Modal dismisses	As expected	PASS
12	Escape key closes modal	Modal dismisses	As expected	PASS
13	Navbar scroll effect	Background solidifies on scroll	As expected	PASS
14	Responsive on mobile	Stacked readable layout	As expected	PASS

Table 6.1: System Testing Results

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The College Portal for Students was successfully designed and implemented as a comprehensive, interactive frontend web application for Rajalakshmi Engineering College. The project achieves all stated objectives including secure authentication, a personalised student dashboard, subject-wise attendance tracking with eligibility indication, internal and semester marks display, course material browsing, and a categorised announcements feed using only HTML5, CSS3, and JavaScript. The application is fully responsive, requires no backend infrastructure, and is freely accessible to students on any modern browser and device.

Through this project, practical experience was gained in frontend web development, UI/UX design principles, DOM manipulation, responsive layout design, and modular code organisation. The portal carries significant academic and social value by providing students with a centralised, organised, and accessible interface for all essential academic information, reducing reliance on physical notice boards and scattered communication channels.

7.2 Future Enhancements

- Backend development using Node.js and Express with a MongoDB or MySQL database for storing and retrieving real student academic data.
- Real-time attendance and marks updates through server-side API integration with the college ERP system.
- Mobile application development using React Native for dedicated Android and iOS platforms with push notification support.
- Push notifications for new announcements, examination schedules, and attendance warnings delivered directly to students.
- Integration of an AI-powered chatbot for answering student queries related to attendance policies, examination schedules, and course information.
- Data visualisation charts using Chart.js for tracking attendance trends and marks progression across semesters.
- Multilingual support to improve accessibility for students across different regional language backgrounds.

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APPENDIX

Appendix 1: System Features Summary

The College Portal for Students includes the following key features:

- Personalised student dashboard with academic profile, summary statistics, timetable, and recent announcements.
- Subject-wise attendance tracking with percentage computation and eligibility status indication.
- Internal assessment and semester examination marks display with grade and GPA computation.
- Course materials module for organised browsing and downloading of subject-specific academic resources.
- Announcements module with categorised and colour-coded notice cards for examinations, events, holidays, and placements.
- Modal-based Sign In and Sign Up authentication with success confirmation.
- Fully responsive layout compatible with mobile, tablet, and desktop screen sizes.

Appendix 2: Test Verification Summary

Test Case 1: Login Functionality

Result: PASS – Valid credentials load the student dashboard. Empty fields trigger an appropriate alert message.

Test Case 2: Attendance Display

Result: PASS – All subjects render with correct attendance percentages and eligibility badges colour-coded appropriately.

Test Case 3: Marks Display

Result: PASS – Internal assessment marks, grades, CGPA, and semester GPA display correctly for all subjects.

Test Case 4: Materials Loading

Result: PASS – Subject cards load with correct resource listings and download links accessible.

Test Case 5: Responsive UI

Result: PASS – Layout adapts correctly from mobile (320px) to desktop (1920px); navigation collapses to hamburger menu on mobile.

Final System Status:

The College Portal for Students has been fully developed and successfully tested. The system provides students of Rajalakshmi Engineering College with centralised access to attendance records, internal marks, course materials, and institutional announcements through a clean, responsive, and interactive frontend web application. The portal is ready for deployment on any standard web hosting environment.

ONLINE COURSE CERTIFICATES: